

Beyond the Position Label: Clustering and Predicting NHL Skater Roles from Per-60 Performance Profiles

Owen McFadden

University of California, San Diego

La Jolla, California, USA

omcfadden@ucsd.edu

Abstract

Position labels in hockey describe where players line up on the ice, but not necessarily how they play. This work asks whether statistical style encodes listed positions, using supervised classification and unsupervised clustering on roughly 94,000 player-game records. Forward vs. defense classification is highly accurate, with logistic regression reaching about 97% accuracy, while the 4-class task performance drops sharply, struggling mainly to distinguish left from right wings. Unsupervised clustering recovers four interpretable forward archetypes (stars, middle-six, fourth-liners, enforcers) without using any labels. Statistical style traces role, not labels. Some position labels align with role while others do not.

Keywords

sports analytics, hockey, clustering, classification, player profiling

1 Introduction

In hockey, each skater is assigned a position that describes where they line up on the ice, but these labels don't always reflect how a player actually plays. There are two general groups, forwards and defensemen. Forwards are further split into centers, left wings, and right wings, with defensemen playing the back of the formation. Position labels offer a rough estimate for playing style, but modern hockey analytics is increasingly interested in measuring role and style directly.

This raises a natural question: does a player's statistical fingerprint encode their listed position, and where might this correspondence break down? Both supervised classification and unsupervised clustering are applied to roughly 94,000 player-game records pulled from the public NHL Stats API, covering the 2022–23 and 2023–24 seasons.

Forward vs. defense classification is highly accurate, with the linear baseline performing at around 97% accuracy, suggesting that the boundary is essentially linear in this feature space. However, the 4-class task performance drops significantly, with the model particularly struggling to distinguish between left and right wings, revealing that wing designation is not a statistically distinct playing style. Without access to labels, K-Means clustering of forward profiles recovers four recognizable archetypes that align with common NHL roles.

Statistical style traces role and contribution, not labels. Some labels, such as defensemen vs. forwards align with role, while others, such as left vs. right wing, don't. The model's failures reveal which labels are statistically real and which are positional conventions. The remainder of this paper describes the data (§2), the prediction task (§3), the methods (§4), the results (§5), and concludes with limitations and future work (§6).

2 Dataset

Data was collected from the NHL Stats API, which provides aggregated box-score counting statistics. Data was pulled team by team to work around the API's pagination limit: the API caps how deep a single query can page at roughly 10,000 rows, so pulling team by team keeps each request below this ceiling.

The player-game dataset contains a record for every player who appeared in a game over the 2022–23 and 2023–24 seasons, totaling roughly 94,000 player-game records. Because playstyle is a per-player, per-season property, the data was aggregated by player and season. Players largely follow a consistent role and style throughout a year but may change season to season. The data was then filtered to retain only player-seasons with 200 or more minutes of ice time, since smaller samples yield noisy rate estimates. After aggregation and filtering, 1,443 player-season profiles remained.

The features were normalized to a per-60-minutes-of-ice-time basis so that player styles could be compared independently of how much ice time each player receives. Star players receive far more ice time than less prominent players, which inflates their raw totals. Normalizing in this way allows players to be compared on style rather than on volume of ice time. The features can be split into offensive and defensive metrics: offensive stats capture scoring (goals, assists, shots), while defensive stats capture physicality and disruption of the opponent's offense (blocks, hits, takeaways, giveaways).

During exploratory data analysis, the most informative visualization was a set of box plots of key per-60 stats broken out by position (C, L, R, D), shown in Figure 1. Several patterns emerged that proved central to comparing player styles. In particular, defensemen shoot, score, and record takeaways far less often than forwards of any type, while recording more blocks and slightly more giveaways. Despite these differences, the most striking pattern was the similarity among C, L, and R, which were nearly identical across all statistics, foreshadowing the difficulty any model would have in distinguishing among forward positions.

3 Predictive Task

Two separate targets were defined: a binary predictive task and a harder 4-class predictive task. The binary task predicts whether a skater is a forward or defenseman based on their statistical profile. Because it separates the broad roles of forward and defenseman, this task was expected to be the simpler of the two. The 4-class task aims to distinguish among the three forward positions and defensemen (C, L, R, D). This task was expected to be considerably more difficult, since C, L, and R share similar values across many box-score statistics.

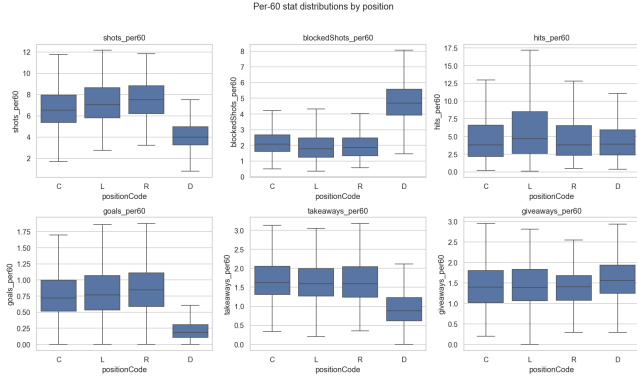


Figure 1: Per-60 distributions of key statistics by listed position. Defensemen separate clearly from forwards (notably in blocked shots and shot volume), while the three forward positions (C, L, R) are nearly indistinguishable.

To evaluate model performance, accuracy and F1 were chosen as metrics. Accuracy is intuitive, as it measures the overall correctness of the model. A common pitfall, however, is that accuracy can become misleading when classes are imbalanced (there are far more C and D than L and R). F1, which combines precision and recall into a single balanced measure, was therefore also used. For the 4-class task, macro-F1 was used so that each position counts equally regardless of how many players fall within each class.

To train and test the models, the data was split by player rather than by player-season. A player with multiple rows (for example, one who played both the 2022–23 and 2023–24 seasons) could otherwise end up in both the training and test sets. Although a player’s style can shift between seasons, players usually follow a consistent archetype. As a result, a player appearing in both sets could allow the model to memorize that player’s profile. The model might then predict the class correctly not because it generalized well, but because it had effectively already seen the player. This would inflate reported accuracy without producing a genuinely better predictor. The training set contains 1,086 player-seasons and the test set 357 player-seasons; the split was verified to contain no overlapping players.

As baselines, logistic regression and Naïve Bayes were used as sensible points of comparison. Logistic regression tests whether the classes are linearly separable, while Naïve Bayes tests whether features are individually informative under an assumption of independence. Both are standard, interpretable starting points.

4 Methods

The main model is gradient boosting, which builds an ensemble of weak learners (typically shallow decision trees), each trained to correct the errors of the previously trained model in order to improve the model’s predictions at each step. Gradient boosting can capture nonlinear interactions between features that the linear baselines cannot. It takes as input the 11 normalized per-60 features described earlier. The scaler was fit on the training data only, to avoid leaking test information into preprocessing.

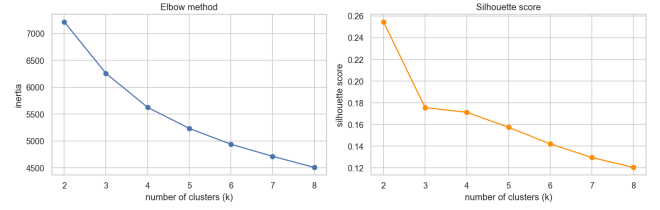


Figure 2: Cluster selection for forward profiles. The elbow plot (left) shows a smooth decline in inertia with no sharp bend, and the silhouette score (right) favors smaller k , indicating that forward styles vary continuously rather than forming sharply separated groups. $k = 4$ was selected as a balance between these criteria and cluster interpretability.

For the unsupervised analysis, K-Means was applied to the forward profiles only, since defensemen already separate clearly from forwards. The more interesting question was whether forwards themselves divide into distinct sub-styles of play.

While K-Means was a natural choice, selecting the number of clusters k required more investigation. The elbow method examines how compact the clusters are, looking for a bend in the plot of within-cluster sum of squares against the number of clusters. The silhouette method instead measures how well separated the clusters are, favoring the k that maximizes cohesion within clusters and separation between them. These selection criteria are shown in Figure 2. Using them, k was set to 4. This was not dictated by the metrics alone, which pointed toward smaller values; instead, $k = 4$ was chosen because it falls within a reasonable range while producing the most interpretable archetypes.

PCA was also used to project the high-dimensional player profiles into a two-dimensional space, purely as a tool for visualizing the style space and clusters (Figure 3), and not as part of the modeling process itself.

5 Results

5.1 Position Classification

The binary task is clearly learnable. Table 1 shows the results of binary classification (forward vs. defense): all three models exceed 87% accuracy, with logistic regression reaching 96.6%. Notably, the linear model performs best of all, which indicates that the two classes are nearly linearly separable. The corresponding confusion matrix (Figure 4) shows a reassuring trend: few players were misclassified. The errors are also roughly symmetric, meaning the model is not systematically biased (for example, it does not simply predict that every player is a forward).

The 4-class task shows a consistent drop in performance, as shown in Table 2, with none of the models exceeding 70% accuracy. As in the binary task, the logistic regression baseline outperforms gradient boosting. The macro-F1 scores drop considerably for all three models, indicating that the models do reasonably well on the larger classes (D, C) but poorly on the smaller ones (L, R). The confusion matrix for the 4-class task (Figure 5) reflects this difficulty. Defensemen are predicted very accurately (115 of 124 correct) and centers reasonably well, but the predictions for L and R scatter

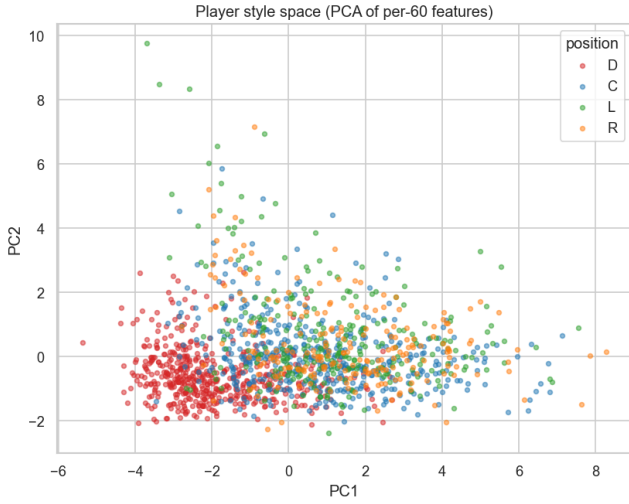


Figure 3: Player profiles projected into two dimensions via PCA, colored by listed position. Defensemen (red) occupy a distinct region from forwards along the first principal component, while the three forward positions intermix heavily.

Table 1: Binary classification results (forward vs. defense).

Model	Accuracy	F1
Logistic Regression	0.966	0.974
Naive Bayes	0.871	0.900
Gradient Boosting	0.933	0.948

Table 2: 4-class classification results (C/L/R/D).

Model	Accuracy	Macro-F1
Logistic Regression	0.700	0.571
Naive Bayes	0.574	0.489
Gradient Boosting	0.616	0.524

heavily into other classes. For example, right wing recall is just 0.18, meaning the model identifies fewer than 1 in 5 actual right wings, compared to an F1 of 0.91 for defensemen. Left and right wings are essentially unpredictable because they do not differ systematically in playstyle, confirming the pattern first seen in the position box plots.

5.2 Ablation Study

To understand which features drive the classifier, the model was tested on each group of features in isolation: all features, scoring-only features, and defensive-only features. As Table 3 shows, removing either group barely hurts performance, dropping from an accuracy of 0.933 with all features to 0.919 using only defensive features. This is because the two sets of features carry overlapping information, so either group alone is sufficient to separate the classes.

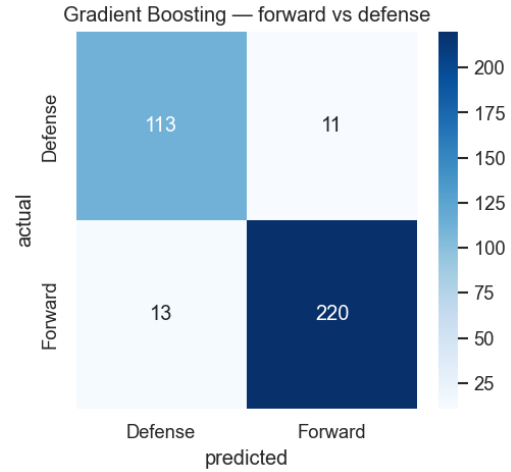


Figure 4: Confusion matrix for the gradient boosting model on the binary task. Misclassifications are rare and roughly symmetric between the two classes.

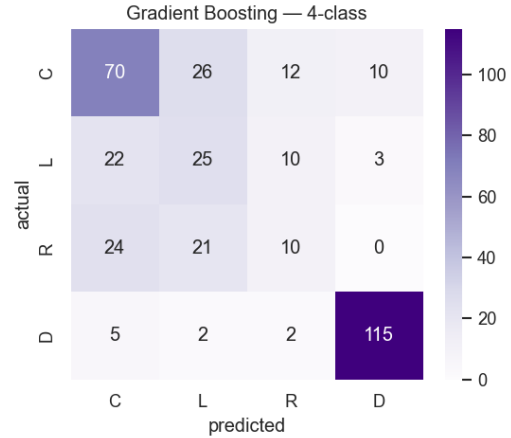


Figure 5: Confusion matrix for the gradient boosting model on the 4-class task. Defensemen are classified accurately and centers reasonably well, but left and right wings scatter heavily into other classes.

Table 3: Ablation study on the binary task (gradient boosting). Performance changes little when either feature group is removed.

Feature Set	Accuracy	F1
All features	0.933	0.948
Scoring only	0.924	0.942
Defensive only	0.919	0.938

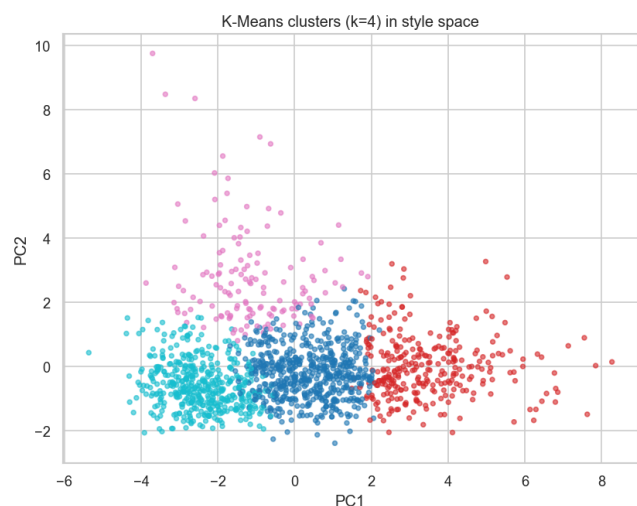


Figure 6: The four K-Means clusters of forward profiles, visualized in the PCA style space.

5.3 Discovered Forward Archetypes

To explore whether forwards divide into meaningful sub-styles, K-Means was applied to the forward profiles with $k = 4$. The resulting clusters, shown in Figure 6, are visually separable in the PCA projection. Despite the fact that the algorithm had no access to position or role labels, the four groups correspond closely to distinct and recognizable forward roles. Figure 7 profiles each cluster by its mean per-60 statistics, and the defining statistics of each style are summarized in Table 4.

Cluster 0 is the middle-six role, the most common forward archetype, represented by players like Eric Staal and Corey Perry. These players are capable of scoring and playmaking, but also show physicality and defensive contribution. Cluster 1 is the star players, including some of the biggest names in hockey like Alex Ovechkin. These players are best known for their scoring (1.21 goals per 60 minutes on average), but also generate large volumes of shots and assists. Cluster 2 is the enforcers, including Ryan Reaves and Milan Lucic. The defining stat for this group is overwhelmingly clear: more than 14 hits per 60 minutes, more than 2.5 times the next-highest cluster. These players are on the ice largely for their physical presence rather than offense. Cluster 3 is the fourth-liners, made up of grinders like Nick Bonino and Casey Cizikas. They fill out the bottom of the lineup as low-event, low-offense checking players whose role is not to fill the stat sheet.

Without access to position or role labels, K-Means reconstructed the actual hierarchy of NHL forwards: stars (cluster 1), middle-six (cluster 0), fourth-liners (cluster 3), and enforcers (cluster 2). The clusters do not correspond to C/L/R, which are difficult to distinguish stylistically; they correspond to roles. Statistical style encodes a player’s role, not their wing designation. The silhouette scores were low and styles vary continuously, so although these clusters represent useful, interpretable groupings, they fall along a continuum rather than perfectly discrete categories. A player near a boundary is a blend of multiple styles.

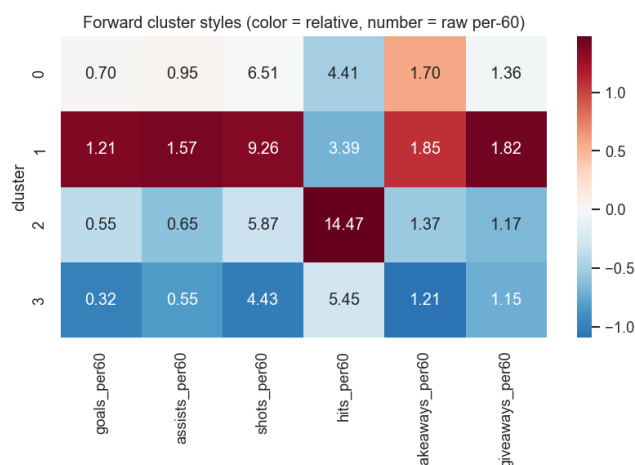


Figure 7: Per-60 statistical profile of each forward cluster. Color shows each stat’s value relative to other clusters (red = high, blue = low), with raw per-60 values overlaid.

Table 4: Defining statistics of the four forward clusters (mean per-60).

Cluster	Goals	Asts	Shots	Hits	n
Middle-six (0)	0.70	0.95	6.51	4.41	477
Stars (1)	1.21	1.57	9.26	3.39	283
Enforcers (2)	0.55	0.65	5.87	14.47	120
Fourth-line (3)	0.32	0.55	4.43	5.45	55

5.4 Stylistic Outliers

Within the binary classification task, 24 out of 357 players in the test data were misclassified, coming in at a 6.7% misclassification rate. This shows that the task was well-learned, and even the players the model misclassifies should not be considered model failures; they identify stylistic and positional mismatches. A forward the model thinks is a defenseman is a forward who plays like a defenseman (physical, blocks shots, fewer shots, etc.). The misclassifications are the model’s way of revealing stylistic outliers.

The first group of players are offensive defensemen, players listed as defensemen whom the model classified as forwards. Some examples of misclassified offensive defensemen are Dougie Hamilton and Aaron Ekblad. These players, along with the other examples, are all classic offensive defensemen, taking a lot of shots and contributing offensively. One stat that makes these players jump out specifically is their shooting, with Hamilton and Ekblad taking 9.24 and 7.80 shots per-60 minutes respectively.

The second group of players are defensive forwards, players listed as forwards whom the model classified as defensemen. Some examples of misclassified defensive forwards are Barclay Goodrow and Luke Glendening. These players are misclassified largely due to their shot-blocking ability, while also scoring at a lower rate than other forwards. For example, Goodrow’s 5.13 blocks and 10.15 hits per-60 minutes match the profile of a defenseman.

Table 5: Selected stylistic outliers from the binary classification task. In each case, the model’s “error” reflects a real divergence between the player’s listed position and their statistical style.

Player	Listed	Predicted	Telltale per-60 stats
Dougie Hamilton	D	Forward	9.24 shots
Aaron Ekblad	D	Forward	7.80 shots
Tony DeAngelo	D	Forward	5.94 shots
Barclay Goodrow	C	Defense	5.13 blocks, 10.15 hits
Luke Glendening	C	Defense	4.26 blocks, 8.10 hits
Jason Dickinson	C	Defense	3.86 blocks, 5.41 hits

An offensive defenseman like Dougie Hamilton is essentially playing a role matching that of a Stars-cluster forward, while a defensive forward like Barclay Goodrow is essentially playing a Fourth-line role. Together, these outliers reinforce the broader finding: statistical styles trace a player’s role and contribution, not their listed position.

6 Related Work

Vincent and Eastman [5], Chan, Cho, and Novati [1], and Säfvenberg, Carlsson, and Lambrix [3] all apply some form of cluster analysis to NHL skater statistics to formally categorize players into styles of play. They argue that data-driven categorization is preferable to ad-hoc labels commonly used by fans and commentators. These span 15 years of NHL position analysis, and all three examine how positional clustering relates to team success and individual performance, but none directly explore how they relate to existing positional labels.

Schulte and Zhao [4] use spatial tracking to cluster and rank NHL players. Rather than just using box-score statistics to analyze a player’s impact, this study uses zone-data, action type, and transition probabilities to quantify skaters’ per-20 minute impact. This kind of richer data source represents a limitation to the present work and is a natural direction for extending it.

Penner [2] uses k-means clustering to identify nine distinct statistical archetypes using data from 110 professional and collegiate basketball leagues. The concept of “positionless” roster composition has been explored deeply in basketball. The present work brings this framing to hockey, where the question of whether listed positions encode distinct playing styles has been less directly examined.

7 Discussion and Conclusion

Statistical style strongly encodes the distinction between forwards and defensemen, but cannot meaningfully encode the distinction between left and right wings. The binary classification task works well, while the 4-class confusion matrix and per-class F1 scores provide direct evidence that L and R are not separable in the feature space we measured. What statistical style does encode well are roles *within* the forward position. Even without supervision, K-Means rediscovered the natural hierarchy of forward roles from raw per-60 statistics alone. While some labels (such as D vs. F) align with playing role, others (such as L vs. R) do not. Statistical style traces role and contribution, making it effective for distinguishing forwards

from defensemen and for identifying roles within forwards, but unable to distinguish positions that do not correspond to a distinct style.

Although the model captures real roles and styles within the sport, there are serious limitations. The most important concerns relate to the data itself: its scope and the noise it contains. The data is limited to box score counting stats, with no play-by-play events, zone or tracking data, or skating speed. As a result, the model measures the style of *what* happens, not *how* it happens. The data also covers only two seasons, so style trajectories over a player’s career and league-wide trends over time are not captured. Finally, the position labels themselves can be noisy: players are sometimes listed at a position they rarely actually play, which adds noise to the supervised task.

Future steps could address several of these limitations. Adding play-by-play and tracking data would allow the model to capture how players play, not just the statistics they accumulate. Gathering data from additional seasons could enable analysis of how a player’s style evolves over their career. Changing the clustering method to GMM (or any other soft-assignment approach) would allow for blended styles, which is a closer match to the continuous style space evident in the silhouette analysis. Together, these results suggest that statistical style can serve as a useful, label-free way to characterize NHL skaters, complementing traditional position labels with a view grounded in how players actually play.

References

- [1] Timothy C. Y. Chan, Justin A. Cho, and David C. Novati. 2012. Quantifying the Contribution of NHL Player Types to Team Performance. *Interfaces* 42, 2 (2012), 131–145. doi:10.1287/inte.1110.0612
- [2] Luke S. J. Penner. 2025. Player Archetypes within Basketball: Optimizing Roster Composition to Create a Championship Team. *Frontiers in Sports and Active Living* 7 (2025), 1639431. doi:10.3389/fspor.2025.1639431
- [3] Rasmus Säfvenberg, Niklas Carlsson, and Patrick Lambrix. 2024. Identifying Player Roles in Ice Hockey. In *Machine Learning and Data Mining for Sports Analytics*, Ulf Brefeld, Jesse Davis, Jan Van Haaren, and Albrecht Zimmermann (Eds.). Springer. doi:10.1007/978-3-031-53833-9_11
- [4] Oliver Schulte and Zeyu Zhao. 2017. Apples-to-Apples: Clustering and Ranking NHL Players Using Location Information and Scoring Impact. In *MIT Sloan Sports Analytics Conference*. Boston, MA.
- [5] Claude B. Vincent and Byron Eastman. 2009. Defining the Style of Play in the NHL: An Application of Cluster Analysis. *Journal of Quantitative Analysis in Sports* 5, 1 (2009). doi:10.2202/1559-0410.1133